Basic Concepts

Computer Organization and Assembly Language

Lec#2

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Next ...

- ❑ Welcome to COAL 2024
- ❑ Assembly-, Machine-, and High-Level Languages
- ❑ Assembly Language Programming Tools
- ❑ Programmer’s View of a Computer System
- ❑ Data Representation

Data Representation

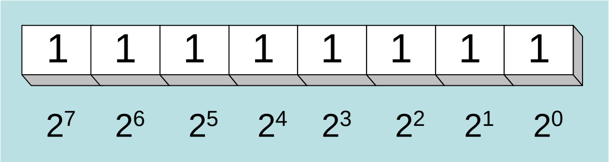
- Binary Numbers
- Hexadecimal Numbers
- Base Conversions
- Integer Storage Sizes
- Binary and Hexadecimal Addition
- Signed Integers and 2's Complement Notation
- Binary and Hexadecimal Subtraction
- Carry and Overflow
- Character Storage

Binary Numbers

- Digits are 1 and 0
  - 1 = true
  - 0 = false
- MSB – most significant bit
- LSB – least significant bit
- Bit numbering:

MSB	LSB
1 0 1 1 0 0 1 0 1 0 0 1 1 1 0 0	
15	0

Binary Numbers



- Each digit (bit) is either 1 or 0
- Each bit represents a power of 2:

Every binary number is a sum of powers of 2

Table 1-3 Binary Bit Position Values.

2^n	Decimal Value	2^n	Decimal Value
2^0	1	2^8	256
2^1	2	2^9	512
2^2	4	2^{10}	1024
2^3	8	2^{11}	2048
2^4	16	2^{12}	4096
2^5	32	2^{13}	8192
2^6	64	2^{14}	16384
2^7	128	2^{15}	32768

Converting Binary to Decimal

Weighted positional notation shows how to calculate the decimal value of each binary bit:

$Decimal = (d_{n-1} \times 2^{n-1}) + (d_{n-2} \times 2^{n-2}) + ... + (d_1 \times 2^1) + (d_0 \times 2^0)$

d = binary digit

binary 00001001 = decimal 9:

$(1 \times 2^3) + (1 \times 2^0) = 9$

Convert Unsigned Decimal to Binary

- Repeatedly divide the decimal integer by 2. Each remainder is a binary digit in the translated value:

Division	Quotient	Remainder
37 / 2	18	1
18 / 2	9	0
9 / 2	4	1
4 / 2	2	0
2 / 2	1	0
1 / 2	0	1

least significant bit

most significant bit

stop when quotient is zero

37 = 100101

Hexadecimal Integers

Binary values are represented in hexadecimal.

Table 1-5 Binary, Decimal, and Hexadecimal Equivalents.

Binary	Decimal	Hexadecimal	Binary	Decimal	Hexadecimal
0000	0	0	1000	8	8
0001	1	1	1001	9	9
0010	2	2	1010	10	A
0011	3	3	1011	11	B
0100	4	4	1100	12	C
0101	5	5	1101	13	D
0110	6	6	1110	14	E
0111	7	7	1111	15	F

Converting Binary to Hexadecimal

- Each hexadecimal digit corresponds to 4 binary bits.
- Example: Translate the binary integer 000101101010011110010100 to hexadecimal:

1	6	A	7	9	4
0001	0110	1010	0111	1001	0100

Converting Hexadecimal to Decimal

- Multiply each digit by its corresponding power of 16:
- *Decimal* = $(d^3 \times 16^3) + (d^2 \times 16^2) + (d^1 \times 16^1) + (d^0 \times 16^0)$
- d = hexadecimal digit
- Examples:
  - Hex 1234 = $(1 \times 16^3) + (2 \times 16^2) + (3 \times 16^1) + (4 \times 16^0)$ =
  - Decimal 4,660
  - Hex 3BA4 = $(3 \times 16^3) + (11 * 16^2) + (10 \times 16^1) + (4 \times 16^0)$ =
  - Decimal 15,268

Converting Decimal to Hexadecimal

- Repeatedly divide the decimal integer by 16. Each remainder is a hex digit in the translated value:

Division	Quotient	Remainder
422 / 16	26	6
26 / 16	1	A
1 / 16	0	1

least significant digit

most significant digit

stop when quotient is zero

Decimal 422 = 1A6 hexadecimal

Integer Storage Sizes

Standard sizes:

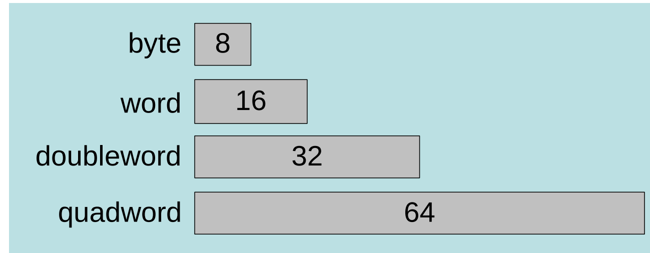


Table 1-4 Ranges of Unsigned Integers.

Storage Type	Range (low–high)	Powers of 2
Unsigned byte	0 to 255	0 to $(2^8 - 1)$
Unsigned word	0 to 65,535	0 to $(2^{16} - 1)$
Unsigned doubleword	0 to 4,294,967,295	0 to $(2^{32} - 1)$
Unsigned quadword	0 to 18,446,744,073,709,551,615	0 to $(2^{64} - 1)$

What is the largest unsigned integer that may be stored in 20 bits?

Binary Addition

- Start with the least significant bit (rightmost bit)
- Add each pair of bits
- Include the carry in the addition, if present

carry:					1				
	0	0	0	0	0	1	0	0	(4)
+	0	0	0	0	0	1	1	1	(7)
	0	0	0	0	1	0	1	1	(11)
bit position:	7	6	5	4	3	2	1	0	

Hexadecimal Addition

- Divide the sum of two digits by the number base (16). The quotient becomes the carry value, and the remainder is the sum digit.

		1	1
36	28	28	6A
42	45	58	4B
78	6D	80	B5

21 / 16 = 1, remainder 5

Important skill: Programmers frequently add and subtract the addresses of variables and instructions.

Signed Integers

- Several ways to represent a signed number
  - Sign-Magnitude
  - Biased
  - 1's complement
  - 2's complement
- Divide the range of values into 2 equal parts
  - First part corresponds to the positive numbers (≥ 0)
  - Second part correspond to the negative numbers (< 0)
- Focus will be on the 2's complement representation
  - Has many advantages over other representations
  - Used widely in processors to represent signed integers

Representation of Negative Numbers

- Is a representation of negative numbers possible?
- Unfortunately:
  - you can not just stick a negative sign in front of a binary number. (it does not work like that)
- There are three methods used to represent negative numbers.
  - Signed magnitude notation
  - Excess notation
  - Two's complement notation

Signed Magnitude Representation

- Unsigned: **-** and **+** are the same.
- In signed magnitude
  - the left-most bit represents the sign of the integer.
    - 0 for positive numbers.
    - 1 for negative numbers.
- The remaining bits represent to magnitude of the numbers.

Example

- Suppose 10011101 is a signed magnitude representation.
- The sign bit is 1, then the number represented is negative

position	7	6	5	4	3	2	1	0
Bit pattern	**1**	**0**	**0**	**1**	**1**	**1**	**0**	**1**
contribution	-			2^4	2^3	2^2		2^0

- The magnitude is 0011101 with a value $2^4+2^3+2^2+2^0 = 29$
- Then the number represented by 10011101 is –29.

Exercise 1

1. 37_{10} has 0010 0101 in signed magnitude notation. Find the signed magnitude of $–37_{10}$?

2. Using the signed magnitude notation find the 8-bit binary representation of the decimal value 24_{10} and -24_{10}.

3. Find the signed magnitude of –63 using 8-bit binary sequence?

Disadvantage of Signed Magnitude

- Addition and subtractions are difficult.
- Signs and magnitude, both have to carry out the required operation.
- They are two representations of 0
  - 00000000 = $+ 0_{10}$
  - 10000000 = $- 0_{10}$
  - To test if a number is 0 or not, the CPU will need to see whether it is 00000000 or 10000000.
  - 0 is always performed in programs.
    - Therefore, having two representations of 0 is inconvenient.

Signed-Summary

- In signed magnitude notation,
  - The most significant bit is used to represent the sign.
  - 1 represents negative numbers
  - 0 represents positive numbers.
  - The unsigned value of the remaining bits represent The magnitude.
- Advantages:
  - Represents positive and negative numbers
- Disadvantages:
  - two representations of zero,
  - Arithmetic operations are difficult.

Excess Representation

– For a given fixed number of bits the range is remapped such that roughly half the numbers are negative and half are positive.

Example: (as left)

Excess 8 -> notation for 4 bit numbers

- **Binary value = 8 + excess-8 value**
- MSB can be used as a sign bit, but
  - If **MSB =1**, positive number
  - If **MSB =0**, negative number
- **Excess Representation is also called *bias***

Excess Value	Notation	Number to store
0	**0000**	-8
1	**0001**	-7
2	**0010**	-6
3	**0011**	-5
4	**0100**	-4
5	**0101**	-3
6	**0110**	-2
7	**0111**	-1
8	**1000**	0
9	**1001**	1
10	**1010**	2
11	**1011**	3
12	**1100**	4
13	**1101**	5
14	**1110**	6
15	**1111**	7

Biased-Excess-N

- Uses a pre-specified number *N* as a biasing value.
- A value is represented by the unsigned number which is *N* greater than the intended value.
  - 0 is represented by *N*,
  - −*N* is represented by the all-zeros bit pattern.

Two's Complement Representation

- Positive numbers
  - Signed value = Unsigned value
- Negative numbers
  - UnSigned value = Signed value + 2^n
  - n = number of bits
- Negative weight for MSB
  - Another way to obtain the signed value is to assign a negative weight to most-significant bit

1	0	1	1	0	1	0	0
-128	64	32	16	8	4	2	1

= -128 + 32 + 16 + 4 = -76

8-bit Binary value	Unsigned value	Signed value
00000000	0	0
00000001	1	+1
00000010	2	+2
. . .	. . .	. . .
01111110	126	+126
01111111	127	+127
10000000	128	-128
10000001	129	-127
. . .	. . .	. . .
11111110	254	-2
11111111	255	-1

Forming the Two's Complement

starting value	**00100100 = +36**
step1: reverse the bits (1's complement)	**11011011**
step 2: add 1 to the value from step 1	**+ 1**
sum = 2's complement representation	**11011100 = -36**

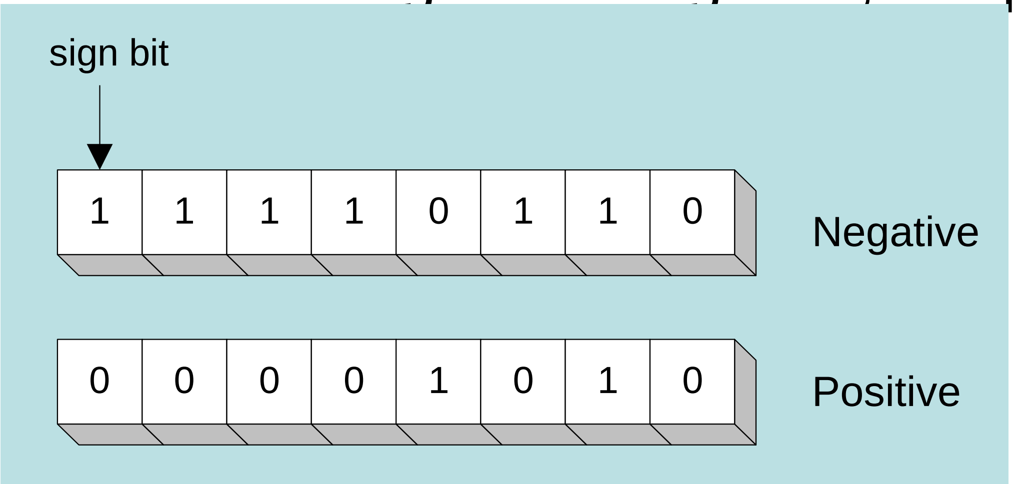
Sum of an integer and its 2's complement must be zero:

00100100 + 11011100 = 00000000 (8-bit sum) $\Rightarrow$ Ignore Carry

The easiest way to obtain the 2's complement of a binary number is by starting at the LSB, leaving all the 0s unchanged, look for the first occurrence of a 1. Leave this 1 unchanged and complement all the bits after it.

Sign Bit

Highest bit indicates the sign. 1 = negative, 0 = positive



If highest digit of a hexadecimal is > 7, the value is negative

Examples: 8A and C5 are negative bytes

A21F and 9D03 are negative words

B1C42A00 is a negative double-word

Sign Extension

Step 1: Move the number into the lower-significant bits

Step 2: Fill all the remaining higher bits with the sign bit

- ❑ This will ensure that both magnitude and sign are correct
- ❑ Examples
  - ➢ Sign-Extend 10110011 to 16 bits

    10110011 = -77 ⇨ **11111111 10110011 = -77**

  - ➢ Sign-Extend 01100010 to 16 bits

    01100010 = +98 ⇨ **00000000 01100010 = +98**
- ❑ Infinite 0s can be added to the left of a positive number
- ❑ Infinite 1s can be added to the left of a negative number

Two's Complement of a Hexadecimal

- To form the two's complement of a hexadecimal
  - Subtract each hexadecimal digit from 15
  - Add 1
- Examples:
  - 2's complement of 6A3D = 95C2 + 1 = 95C3
  - 2's complement of 92F0 = 6D0F + 1 = 6D10
  - 2's complement of FFFF = 0000 + 1 = 0001
- No need to convert hexadecimal to binary

Binary Subtraction

- When subtracting A – B, convert B to its 2's complement
- Add A to (–B)

	0 0 0 0 1 1 0 0	→		0 0 0 0 1 1 0 0
–	0 0 0 0 0 0 1 0		+	1 1 1 1 1 1 1 0 (2's complement)
				0 0 0 0 1 0 1 0 (same result)

- Carry is ignored, because
  - Negative number is sign-extended with 1's
  - You can imagine infinite 1's to the left of a negative number
  - Adding the carry to the extended 1's produces extended zeros

Practice: Subtract 00100101 from 01101001.

Hexadecimal Subtraction

- When a borrow is required from the digit to the left, add 16 (decimal) to the current digit's value

16 + 5 = 21

```
     -1
   C675
-  A247
   ----
   242E
```

```
   1 1
   C675
+  5DB9   (2's complement)
   ----
   242E   (same result)
```

- Last Carry is ignored

Practice: The address of **var1** is 00400B20. The address of the next variable after var1 is 0040A06C. How many bytes are used by var1?

Ranges of Signed Integers

The unsigned range is divided into two signed ranges for positive and negative numbers

Storage Type	Range (low–high)	Powers of 2
Signed byte	−128 to +127	-2^7 to $(2^7 - 1)$
Signed word	−32,768 to +32,767	-2^{15} to $(2^{15} - 1)$
Signed doubleword	−2,147,483,648 to 2,147,483,647	-2^{31} to $(2^{31} - 1)$
Signed quadword	−9,223,372,036,854,775,808 to +9,223,372,036,854,775,807	-2^{63} to $(2^{63} - 1)$

Practice: What is the range of signed values that may be stored in 20 bits?

Carry and Overflow

- Carry is important when …
  - Adding or subtracting unsigned integers
  - Indicates that the unsigned sum is out of range
  - Either < 0 or >maximum unsigned *n*-bit value
- Overflow is important when …
  - Adding or subtracting signed integers
  - Indicates that the signed sum is out of range
- Overflow occurs when
  - Adding two positive numbers and the sum is negative
  - Adding two negative numbers and the sum is positive
  - Can happen because of the fixed number of sum bits

Carry and Overflow Examples

- We can have carry without overflow and vice-versa
- Four cases are possible

				1					
	0	0	0	0	1	1	1	1	15
+	0	0	0	0	1	0	0	0	8
	0	0	0	1	0	1	1	1	23

Carry = 0 Overflow = 0

1	1	1	1	1					
	0	0	0	0	1	1	1	1	15
+	1	1	1	1	1	0	0	0	248 (-8)
	0	0	0	0	0	1	1	1	263 (+7)

Carry = 1 Overflow = 0

	1								
	0	1	0	0	1	1	1	1	79
+	0	1	0	0	0	0	0	0	64
	1	0	0	0	1	1	1	1	143 (-113)

Carry = 0 Overflow = 1

1			1	1					
	1	1	0	1	1	0	1	0	218 (-38)
+	1	0	0	1	1	1	0	1	157 (-99)
	0	1	1	1	0	1	1	1	375 (-137)

Carry = 1 Overflow = 1

Character Storage

- Character sets
  - Standard ASCII: 7-bit character codes (0 – 127)
  - Extended ASCII: 8-bit character codes (0 – 255)
  - Unicode: 16-bit character codes (0 – 65,535)
  - Unicode standard represents a universal character set
    - Defines codes for characters used in all major languages
    - Used in Windows-XP: each character is encoded as 16 bits
  - UTF-8: variable-length encoding used in HTML
    - Encodes all Unicode characters
    - Uses 1 byte for ASCII, but multiple bytes for other characters
- Null-terminated String
  - Array of characters followed by a NULL character

Printable ASCII Codes

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
2	space	!	"	#	$	%	&	'	(	)	*	+	,	-	.	/
3	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7	p	q	r	s	t	u	v	w	x	y	z	{	|	}	~	DEL

- Examples:
  - ASCII code for space character = 20 (hex) = 32 (decimal)
  - ASCII code for 'L' = 4C (hex) = 76 (decimal)
  - ASCII code for 'a' = 61 (hex) = 97 (decimal)

Control Characters

- The first 32 characters of ASCII table are used for control
- Control character codes = 00 to 1F (hex)
  - Not shown in previous slide
- Examples of Control Characters
  - Character 0 is the NULL character $\Rightarrow$ used to terminate a string
  - Character 9 is the Horizontal Tab (HT) character
  - Character 0A (hex) = 10 (decimal) is the Line Feed (LF)
  - Character 0D (hex) = 13 (decimal) is the Carriage Return (CR)
  - The LF and CR characters are used together
    - They advance the cursor to the beginning of next line
- One control character appears at end of ASCII table
  - Character 7F (hex) is the Delete (DEL) character

Terminology for Data Representation

- Binary Integer
  - Integer stored in memory in its binary format
  - Ready to be used in binary calculations
- ASCII Digit String
  - A string of ASCII digits, such as "123"
- ASCII binary
  - String of binary digits: "01010101"
- ASCII decimal
  - String of decimal digits: "6517"
- ASCII hexadecimal
  - String of hexadecimal digits: "9C7B"

Summary

- Assembly language helps you learn how software is constructed at the lowest levels
- Assembly language has a one-to-one relationship with machine language
- An assembler is a program that converts assembly language programs into machine language
- A linker combines individual files created by an assembler into a single executable file
- A debugger provides a way for a programmer to trace the execution of a program and examine the contents of memory and registers
- A computer system can be viewed as consisting of layers. Programs at one layer are translated or interpreted by the next lower-level layer
- Binary and Hexadecimal numbers are essential for programmers working at the machine level.